

RANK-TWO CASES OF THE ORDER-FOUR SOULES PERMANENT-ON-TOP PROBLEM

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ABSTRACT. For a positive semidefinite matrix A of order four, Soules' problem asks whether the largest eigenvalue of its Schur power matrix is at most $\text{per}(A)$. We prove this inequality for every real positive semidefinite matrix of rank at most two. For complex matrices of rank at most two, we prove the complete inequality on the irreducible $(2, 2)$ block and a sharp trace bound on the remaining $(3, 1)$ block. The latter is proved by a Bloch-sphere reduction and an elementary three-variable inequality; equality occurs precisely at a regular tetrahedron of Bloch vectors. Consequently, the unresolved complex rank-two case reduces to one explicit determinant inequality for a 3×3 positive semidefinite block. We also settle the full complex rank-two assertion for coplanar Bloch configurations and for a non-coplanar cyclic family. All algebraic identities used in the proof are amenable to exact symbolic verification. We do not claim a resolution of the unrestricted complex order-four problem.

1. INTRODUCTION

For $A = (a_{ij}) \in \mathbb{C}^{n \times n}$, its Schur power matrix is indexed by $S_n \times S_n$ and is defined by

$$(1) \quad \Pi(A)_{\sigma, \tau} = \prod_{j=1}^n a_{\sigma(j), \tau(j)}.$$

If A is Hermitian positive semidefinite, so is $\Pi(A)$. Its row sums are $\text{per}(A)$, hence the all-ones vector is an eigenvector with eigenvalue $\text{per}(A)$. Soules conjectured that this eigenvalue is the largest one [3]:

$$(2) \quad \lambda_{\max}(\Pi(A)) \leq \text{per}(A).$$

The assertion is known in orders at most three [1]. It fails in order five, even for complex positive semidefinite matrices of rank two [2, 4]. The order-four problem is therefore a natural remaining finite-dimensional case; see [5] for the historical context.

The purpose of this paper is to isolate and prove substantial rank-two parts of the order-four problem. The first statement is a complete theorem

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over the reals. The second and third statements identify what can be proved for arbitrary complex rank-two matrices and reduce the rest to one scalar inequality.

Theorem 1.1. *Let A be a Hermitian positive semidefinite matrix of order four with $\text{rank}(A) \leq 2$.*

- (1) *If A is real, then (2) holds.*
- (2) *For arbitrary complex A , every eigenvalue belonging to the irreducible $(2, 2)$ block of $\Pi(A)$ is at most $\text{per}(A)$.*
- (3) *If B_{31} denotes the positive semidefinite 3×3 matrix representing the nonzero spectrum of the $(3, 1)$ block and $P = \text{per}(A)$, then*

$$(3) \quad \text{tr } B_{31} \leq 2P.$$

The constant 2 is sharp. Equality occurs precisely when the four associated Bloch vectors form a regular tetrahedron.

Consequently, the complex rank-two case is equivalent to the single remaining condition

$$(4) \quad \det(PI_3 - B_{31}) \geq 0.$$

The paper is organized as follows. Section 2 performs the rank-two and representation-theoretic reductions. Section 3 settles the complex $(2, 2)$ block. Section 4 proves the real $(3, 1)$ block by an explicit sum of squares. Section 5 proves the sharp complex trace theorem. Section 6 records two complex classes for which the full rank-two assertion follows.

2. RANK-TWO AND FOURIER-BLOCK REDUCTIONS

We first record two invariances. If D is diagonal, then $\text{per}(DAD^*) = |\det D|^2 \text{per}(A)$ and $\Pi(DAD^*) = |\det D|^2 \Pi(A)$. Diagonal unitary congruence therefore preserves both sides of (2), and positive diagonal congruence scales them by the same factor. Zero diagonal entries of a positive semidefinite matrix give zero rows and columns and cause no difficulty.

On a dense set, a rank-two Gram matrix is consequently diagonally congruent to

$$(5) \quad H_{ij} = 1 + x_i \overline{x_j}, \quad x_i \in \mathbb{C}.$$

The boundary follows by continuity. Let $e_k = e_k(x_1, x_2, x_3, x_4)$ be the elementary symmetric functions, with $e_0 = 1$. The permanent has the standard rank-two expansion

$$(6) \quad P := \text{per}(H) = \sum_{k=0}^4 k!(4-k)! |e_k|^2.$$

For a matrix G of order four, define its first associated matrix by

$$(7) \quad C_1(G)_{ij} = g_{ij} \text{ per } G(i \mid j),$$

where $G(i \mid j)$ is obtained by deleting row i and column j .

Lemma 2.1. *The only nonzero Fourier blocks of $\Pi(H)$ are indexed by the partitions*

$$(4), \quad (3, 1), \quad (2, 2).$$

The (4) block is P . The spectrum of the (3, 1) block is the spectrum of $C_1(H)$ on $\mathbf{1}^\perp$. Moreover, for $k = 1, 2, 3$, if

$$(8) \quad v_k(i) = 4e_k(x_1, \dots, \hat{x}_i, \dots, x_4) - (4 - k)e_k,$$

then $v_k \perp \mathbf{1}$ and

$$(9) \quad C_1(H) = \frac{P}{4}\mathbf{1}\mathbf{1}^* + \frac{1}{2}v_1v_1^* + \frac{1}{4}v_2v_2^* + \frac{1}{2}v_3v_3^*.$$

Proof. Put $f(\sigma) = \prod_i h_{i, \sigma(i)}$. The Schur power matrix is the group-circulant matrix of f . Its Fourier transform in the permutation representation of S_4 is, up to transpose, $C_1(H)$: indeed, its (i, j) entry is the sum of $f(\sigma)$ over permutations satisfying $\sigma(j) = i$, which is h_{ji} per $H(j \mid i)$. Since the permutation representation is the direct sum of the trivial and standard representations, $C_1(H)$ contains the (4) and (3, 1) blocks. Its row sums equal P , so the trivial block is P .

Schur–Weyl duality for $(\mathbb{C}^2)^{\otimes 4}$ eliminates the partitions with more than two rows, leaving only (4), (3, 1), (2, 2). Finally, expand $h_{ij} = 1 + x_i\bar{x}_j$ in (7) and group terms by their degree in x . The result is (9); summing (8) over i shows $v_k \perp \mathbf{1}$. $\square$

Thus, on $\mathbf{1}^\perp$, the (3, 1) block has the same nonzero eigenvalues as

$$F = \frac{1}{2}v_1v_1^* + \frac{1}{4}v_2v_2^* + \frac{1}{2}v_3v_3^*.$$

Equivalently, if $Z = [v_1 \ v_2 \ v_3]$ and $D = \text{diag}(2^{-1/2}, 2^{-1}, 2^{-1/2})$, we may take

$$(10) \quad B_{31} = DZ^*ZD.$$

3. THE COMPLEX (2, 2) BLOCK

Define the three Plücker-type expressions

$$(11) \quad \begin{aligned} q_1 &= (x_2 - x_1)(x_4 - x_3), \\ q_2 &= (x_3 - x_1)(x_4 - x_2), \\ q_3 &= (x_4 - x_1)(x_3 - x_2). \end{aligned}$$

Lemma 3.1. *The (2, 2) block has rank at most one. Its possible nonzero eigenvalue is*

$$(12) \quad T_{22} = 2 \sum_{j=1}^3 |q_j|^2.$$

Proof. The (2, 2) representation occurs with multiplicity one in $(\mathbb{C}^2)^{\otimes 4}$, so the corresponding positive semidefinite Fourier block has rank at most one. Applying the (2, 2) character projector and taking its trace gives (12). This is also obtained directly by expanding the three standard polytabloid

coordinates, which are precisely the expressions in (11) up to a fixed unitary change of basis. $\square$

Proposition 3.2. *For arbitrary $x_1, \dots, x_4 \in \mathbb{C}$, one has $T_{22} \leq P$.*

Proof. A common unitary transformation of the two-dimensional Gram vectors induces a Möbius change of the affine coordinates and changes (5) only by diagonal congruence. We may therefore send the first projective point to 0. By continuity, it is enough to put $x_1 = 0$ and write $(x_2, x_3, x_4) = (a, b, c)$. Direct expansion gives

$$(13) \quad \frac{P - T_{22}}{6} = 4 + |a + b + c|^2 + |ab + ac + bc|^2 + |abc|^2 - |ab|^2 - |ac|^2 - |bc|^2.$$

As a quadratic polynomial in c and $\bar{c}$, the right-hand side is

$$(14) \quad \alpha |c|^2 + 2 \operatorname{Re}(\beta c) + \gamma,$$

where

$$(15) \quad \alpha = |1 + \bar{a}b|^2, \quad \beta = \bar{a} + \bar{b} + \bar{a}\bar{b}(a + b), \quad \gamma = 4 + |a + b|^2.$$

The associated Hermitian 2×2 coefficient matrix is positive semidefinite, since

$$(16) \quad \alpha\gamma - |\beta|^2 = 4(1 + \operatorname{Re}(\bar{a}b))^2 \geq 0.$$

If $\alpha = 0$, the same identity forces $\beta = 0$, while $\gamma > 0$. Thus (14) is nonnegative in every case. $\square$

4. THE REAL RANK-TWO THEOREM

Assume now that the coordinates in the preceding chart are real, and set

$$(17) \quad p = a + b + c, \quad q = ab + ac + bc, \quad r = abc.$$

By (10), the inequality $B_{31} \preceq PI_3$ is equivalent to

$$(18) \quad M := \operatorname{diag}(2P, 4P, 2P) - Z^T Z \succeq 0.$$

Exact simplification gives $M = 4N$, where

$$(19) \quad N = \begin{pmatrix} 2q^2 + 8q + 3r^2 + 12 & -2(pq - 6r) & -pr \\ -2(pq - 6r) & 2(3p^2 + 4pr + 3r^2 + 12) & -2qr \\ -pr & -2qr & 3p^2 + 2q^2 + 12 \end{pmatrix}.$$

Proposition 4.1. *The matrix N in (19) is positive semidefinite for all $p, q, r \in \mathbb{R}$.*

Proof. For $u, v, w \in \mathbb{R}$, a direct expansion yields the exact identity

$$(20) \quad \begin{aligned} (u, v, w)N(u, v, w)^T &= \frac{4}{3}((3 + q)u + rv)^2 + \frac{2}{3}(qu - (3p + 2r)v)^2 \\ &\quad + 2(qw - rv)^2 + \frac{1}{3}(3ru + 8v - pw)^2 \\ &\quad + \frac{8}{3}(v + pw)^2 + 12w^2. \end{aligned}$$

Every term on the right is nonnegative. $\square$

Proposition 4.1 proves the $(3, 1)$ block inequality. Proposition 3.2 proves the remaining nontrivial block inequality. This establishes Theorem 1.1(1). In the same real chart the $(2, 2)$ defect has the particularly simple strict form

$$(21) \quad P - T_{22} = 6((p + r)^2 + 4) > 0.$$

5. A SHARP COMPLEX TRACE BOUND

We turn to arbitrary complex rank-two matrices. By positive diagonal congruence, normalize the four Gram vectors to unit spinors. Let a_{ij} denote their Gram matrix, put $s = \text{tr } B_{31}$, and set

$$(22) \quad c = \sum_{1 \leq i < j \leq 4} |a_{ij}|^2.$$

Lemma 5.1. *One has*

$$(23) \quad c = \frac{P}{4} + \frac{s}{4}.$$

Proof. Let $w = u_1 \otimes \cdots \otimes u_4$, where the u_i are the four unit spinors, and let E_λ be the squared norm of the projection of w onto the λ -isotypic component. If d_λ is the degree of λ , the character projector gives

$$\text{tr } B_\lambda = \frac{24}{d_\lambda} E_\lambda.$$

In particular, $E_{(4)} = P/24$ and $E_{(3,1)} = s/8$. The class sum of the six transpositions acts by the scalars $6, 2, 0$ on the irreducible representations $(4), (3, 1), (2, 2)$, respectively. Its expectation in w is

$$\sum_{i < j} |\langle u_i, u_j \rangle|^2 = c.$$

Consequently $c = 6E_{(4)} + 2E_{(3,1)} = P/4 + s/4$. $\square$

Associate to each unit spinor its Bloch vector $n_i \in S^2 \subset \mathbb{R}^3$; then $|a_{ij}|^2 = (1 + n_i \cdot n_j)/2$. Define

$$(24) \quad \begin{aligned} S &= \sum_{i < j} n_i \cdot n_j, \\ Q &= (n_1 \cdot n_2)(n_3 \cdot n_4) + (n_1 \cdot n_3)(n_2 \cdot n_4) + (n_1 \cdot n_4)(n_2 \cdot n_3). \end{aligned}$$

Lemma 5.2. *With the preceding notation,*

$$(25) \quad P = \frac{15}{2} + \frac{5}{2}S + \frac{1}{2}Q, \quad c = 3 + \frac{1}{2}S.$$

Proof. The second identity follows immediately from $|a_{ij}|^2 = (1 + n_i \cdot n_j)/2$. For the first, use the coherent-state resolution of the identity on $\text{Sym}^4(\mathbb{C}^2)$,

$$P_{\text{sym}} = 5 \int |\psi^{\otimes 4}\rangle \langle \psi^{\otimes 4}| d\psi.$$

Since $\langle w, P_{\text{sym}} w \rangle = P/24$ and $|\langle \psi, u_i \rangle|^2 = (1 + m \cdot n_i)/2$, where m is the Bloch vector of ψ , we obtain

$$P = \frac{120}{16} \int_{S^2} \prod_{i=1}^4 (1 + m \cdot n_i) dm.$$

The odd moments vanish. Applying the identities $\int u_i u_j du = \delta_{ij}/3$ and $\int u_i u_j u_k u_l du = (\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})/15$ gives (25). $\square$

Proposition 5.3. *For four unit vectors $n_1, \dots, n_4 \in \mathbb{R}^3$,*

$$(26) \quad 21 + 11S + 3Q \geq 0.$$

Equality holds if and only if the four vectors form a regular tetrahedron.

Proof. Fix three vectors and call them a, b, c . Put

$$x = a \cdot b, \quad y = a \cdot c, \quad z = b \cdot c.$$

If d is the fourth unit vector, the left-hand side of (26) is $K + d \cdot w$, where

$$(27) \quad K = 21 + 11(x + y + z), \quad w = (11 + 3z)a + (11 + 3y)b + (11 + 3x)c.$$

The Gram determinant condition gives $x + y + z \geq -3/2$, hence $K \geq 9/2$. Direct expansion gives

$$(28) \quad K^2 - \|w\|^2 = 2E,$$

where, after setting $X = x + 1/3$, $Y = y + 1/3$, $Z = z + 1/3$,

$$(29) \quad E = 24(X^2 + Y^2 + Z^2) + 32(X + Y + Z)^2 - 27XYZ.$$

If $XYZ \leq 0$, then $E \geq 0$. If $XYZ > 0$, put $g = (XYZ)^{1/3}$. Since $X, Y, Z \in [-2/3, 4/3]$, the arithmetic–geometric mean inequality gives

$$E \geq 72g^2 - 27g^3 = 9g^2(8 - 3g) \geq 0.$$

Thus $K \geq \|w\|$, and $K + d \cdot w \geq 0$.

If $XYZ \leq 0$, equality in (29) forces $X = Y = Z = 0$, because its quadratic part is positive definite and $-27XYZ \geq 0$. If $XYZ > 0$, then $0 < g \leq 4/3$, so the displayed lower bound for E is strict. Hence equality is possible only when $x = y = z = -1/3$. In that case $K = 10$, $w = 10(a + b + c)$, and $\|a + b + c\| = 1$; equality in $K + d \cdot w \geq 0$ is therefore equivalent to $d = -(a + b + c)$. This is exactly the regular tetrahedral configuration. The converse is immediate. $\square$

We can now prove Theorem 1.1(3). By Lemma 5.2, inequality (26) is equivalent to $P \geq 4c/3$. Lemma 5.1 then gives $s = 4c - P \leq 2P$. The equality statement follows from Proposition 5.3.

Finally, B_{31} is positive semidefinite. The trace bound therefore allows at most one of its three eigenvalues to exceed P . If one does, the other two are strictly smaller than P , again by the trace bound, so $\det(PI_3 - B_{31}) < 0$. Conversely, if none does, the determinant is nonnegative. Hence $B_{31} \preceq PI_3$ is equivalent to (4). This proves the last assertion of Theorem 1.1 and identifies precisely the remaining obstacle in the complex rank-two problem.

6. TWO COMPLEX CLASSES

Corollary 6.1. *Suppose the four Bloch vectors associated with a complex rank-two Gram matrix lie in a plane through the origin. Then (2) holds.*

Proof. A common unitary rotates the plane to the xz -plane. Individual spinor phases then make all four Gram vectors real. Diagonal unitary congruence preserves the spectral comparison, so Theorem 1.1(1) applies. $\square$

Proposition 6.2. *For $t \geq 0$, consider the complex rank-two chart*

$$(30) \quad (x_1, x_2, x_3, x_4) = (t, it, -t, -it).$$

Then (2) holds strictly on every nontrivial Fourier block.

Proof. Here $e_1 = e_2 = e_3 = 0$ and $e_4 = -t^4$, so $P = 24(1 + t^8)$. The three frame vectors in (9) are mutually orthogonal, and the $(3, 1)$ eigenvalues are

$$32t^2, \quad 16t^4, \quad 32t^6.$$

Writing $y = t^2$, the three defects are

$$\begin{aligned} P - 32t^2 &= 8(3y^4 - 4y + 3), \\ P - 16t^4 &= 8(3y^4 - 2y^2 + 3), \\ P - 32t^6 &= 8(3y^4 - 4y^3 + 3). \end{aligned}$$

The first polynomial is positive because its unique positive critical point satisfies $y^3 = 1/3$, where its value is $3(1 - y) > 0$; the third follows by reciprocity, and the middle one has negative discriminant after setting $z = y^2$. Proposition 3.2 handles the $(2, 2)$ block. $\square$

7. EXACT VERIFICATION AND THE REMAINING PROBLEM

The identities (6), (12), (13), (16), (19), (20), (23), (25), (29), and the cyclic-family formulas have been expanded independently in exact symbolic arithmetic. No floating-point computation is used in any proof above.

The principal unresolved question is whether (4) holds for all complex rank-two configurations. The trace theorem is sharp and cannot be strengthened merely by replacing $2P$ with a smaller universal trace bound. Thus a complete proof must exploit the determinant or, equivalently, additional information about the distribution of the three $(3, 1)$ eigenvalues. The unrestricted complex order-four conjecture also includes ranks three and four and is not addressed by the present theorem.

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REFERENCES

1. R. B. Bapat and V. S. Sunder, *An extremal property of the permanent and the determinant*, Linear Algebra and its Applications **76** (1986), 153–163.
2. V. S. Shchesnovich, *The permanent-on-top conjecture is false*, Linear Algebra and its Applications **490** (2016), 196–201.
3. George W. Soules, *An approach to the permanental-dominance conjecture*, Linear Algebra and its Applications **201** (1994), 211–229.
4. Hoang Anh Tran, *A simple counterexample for the permanent-on-top conjecture*, Mathematical Inequalities & Applications **25** (2022), no. 1, 1–16.
5. Fuzhen Zhang, *An update on a few permanent conjectures*, Special Matrices **4** (2016), no. 1, 305–316.

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